PYTHON INDXING & SLICING

In [1]:
```
s1 = 'welcometeam'
s1
```

Out[1]:
```
'welcometeam'
```

In [2]:
```
type(s1)
```

Out[2]:
```
str
```

In [3]:
```
len(s1)
```

Out[3]:
```
11
```

In [4]:
```
s1
```

Out[4]:
```
'welcometeam'
```

In [5]:
```
s1[1] # forward index
```

Out[5]:
```
'e'
```

In [6]:
```
s1
```

Out[6]:
```
'welcometeam'
```

In [7]:
```
s1[-1]
```

Out[7]:
```
'm'
```

In [8]:
```
s1
```

Out[8]:
```
'welcometeam'
```

In [9]:
```
s1[:]
```

Out[9]:
```
'welcometeam'
```

In [10]:
```
s1[7:11]
```

Out[10]:
```
'team'
```

In [11]:
```
s1
```

Out[11]:
```
'welcometeam'
```

In [12]:
```
s1[7:]
```

Out[12]: 'team'

In [13]:
```
s1[:]
```

Out[13]: 'welcometeam'

In [14]:
```
s1[2:10]
```

Out[14]: 'lcometea'

In [15]:
```
s1
```

Out[15]: 'welcometeam'

In [16]:
```
s1[:10]
```

Out[16]: 'welcometea'

In [17]:
```
s1
```

Out[17]: 'welcometeam'

In [18]:
```
s1[4:]
```

Out[18]: 'ometeam'

In [19]:
```
s1
```

Out[19]: 'welcometeam'

In [20]:
```
s1[12:]
```

Out[20]: ''

In [21]:
```
s1[13]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[21], line 1
----> 1 s1[13]

IndexError: string index out of range
```

In [22]:
```
s1[:13]
```

Out[22]: 'welcometeam'

-s1[left:right] -s1[left:right:step]

In [23]:
```
s1
```

Out[23]: 'welcometeam'

In [24]:
```
s1[:15]
```
Out[24]:
```
'welcometeam'
```

In [25]:
```
s1
```
Out[25]:
```
'welcometeam'
```

In [26]:
```
s1[1:8:2]
```
Out[26]:
```
'ecmt'
```

In [27]:
```
s1
```
Out[27]:
```
'welcometeam'
```

In [28]:
```
s1[::3]
```
Out[28]:
```
'wcea'
```

In [29]:
```
s1
```
Out[29]:
```
'welcometeam'
```

In [30]:
```
s1[-9:-1:3]
```
Out[30]:
```
'lme'
```

In [31]:
```
s1
```
Out[31]:
```
'welcometeam'
```

In [32]:
```
s1[::-1]
```
Out[32]:
```
'maetemoclew'
```

In [33]:
```
s1
```
Out[33]:
```
'welcometeam'
```

In [34]:
```
s1[::-2]
```
Out[34]:
```
'meeolw'
```

In [ ]:
```

```

NUMBER SYSTEM

In [35]:
```
25
```

Out[35]: 25

In [36]:
```
bin(25)
```

Out[36]: '0b11001'

In [37]:
```
int(0b11001)
```

Out[37]: 25

In [38]:
```
25
```

Out[38]: 25

In [39]:
```
oct(25)
```

Out[39]: '0o31'

In [40]:
```
int(0o31)
```

Out[40]: 25

In [41]:
```
hex(0)
```

Out[41]: '0x0'

In [42]:
```
hex(10)
```

Out[42]: '0xa'

In [43]:
```
hex(11)
```

Out[43]: '0xb'

In [44]:
```
hex(12)
```

Out[44]: '0xc'

In [45]:
```
hex(13)
```

Out[45]: '0xd'

In [46]:
```
hex(14)
```

Out[46]: '0xe'

In [47]:
```
hex(15)
```

Out[47]: '0xf'

In [48]:
```
hex(16)
```

Out[48]: '0x10'

In [49]:
```
int(0x10)
```

Out[49]: 16

In [ ]:

BITWISE OPERATOR

In [50]:
```
# complement
~11
```

Out[50]: -12

In [51]:
```
~34
```

Out[51]: -35

In [52]:
```
~23
```

Out[52]: -24

In [ ]:
```
## Bitwise operators
1. &(AND) --> 1 if both bits are 1
2. I(OR) --> 1 if at least one bit is 1
3. (XOR) -->  1 if bits are different
4. (NOT) --> flips bits
5. << (Left Shift) --> shifts bits left
6. >> (Right Shift) --> shifts bits right
```

In [ ]:

In [ ]:

In [ ]:

In [ ]: